

Western Mathematics

TOPIC REVISION QUESTIONS

Mathematics Standard 2

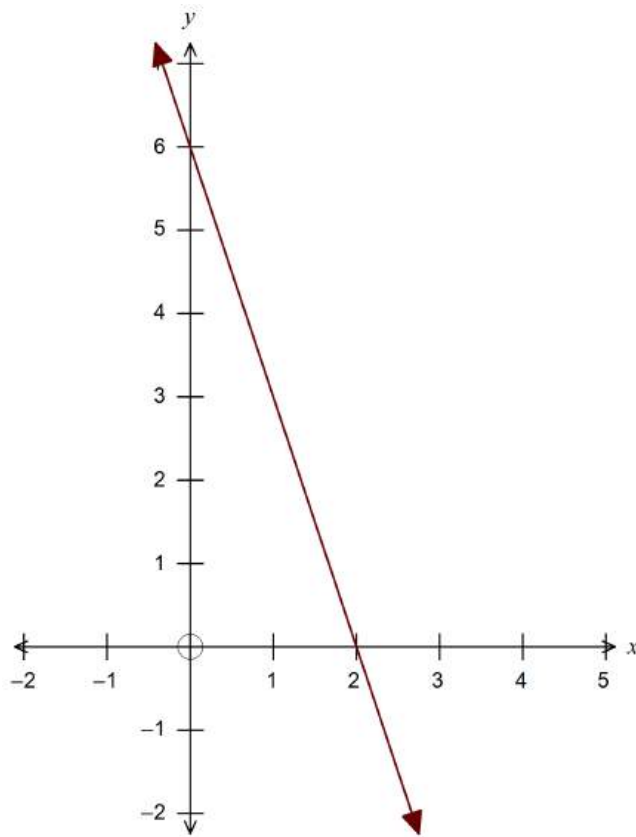
MS-A2 Linear Relationships

Section I – Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

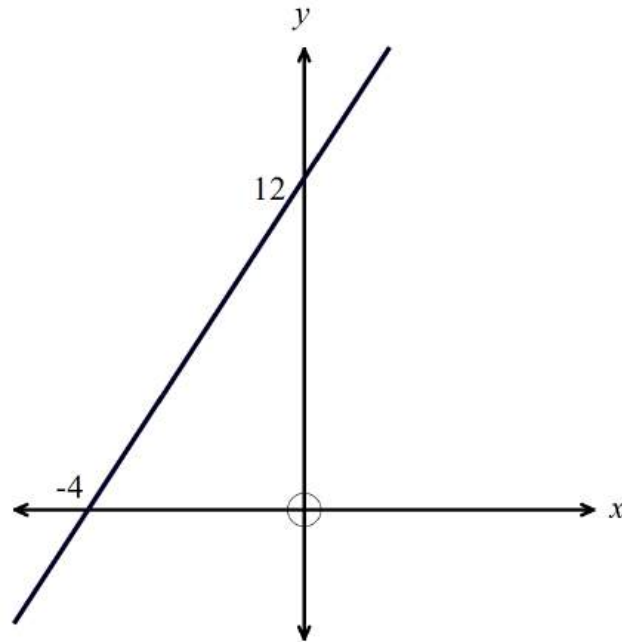
Section I**Multiple Choice Questions (1 Mark each)****10 marks**

1. In the diagram below, what is the gradient of the line shown?



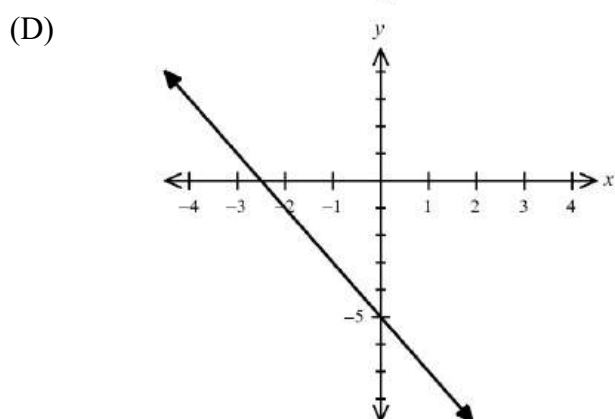
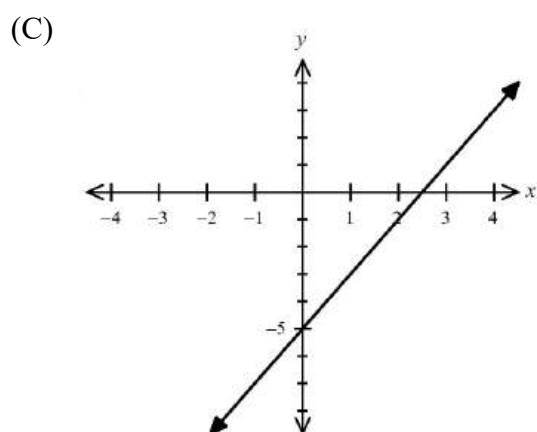
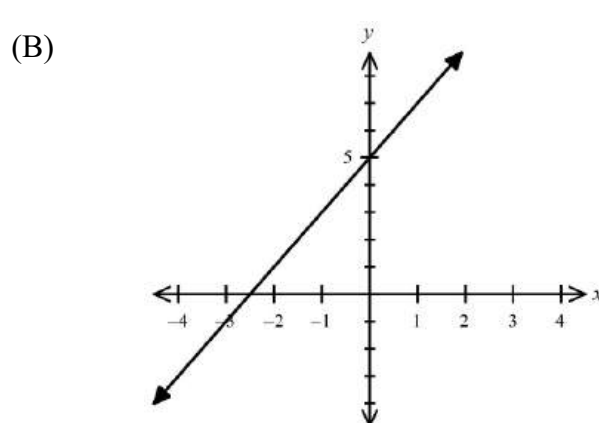
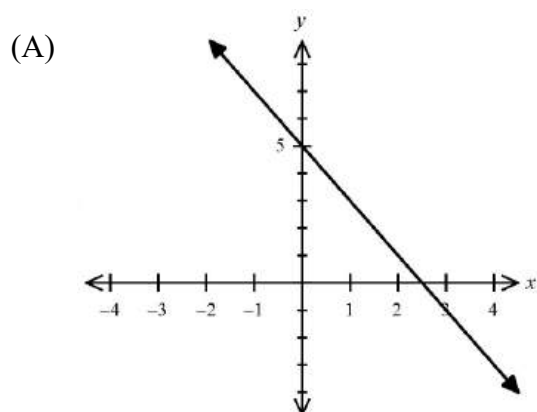
- (A) -6
- (B) -3
- (C) 2
- (D) 3

2. What is the gradient of the line shown below?



- (A) -3
(B) -2
(C) 3
(D) 4
3. A straight line has its equation $x - 2y + 6 = 0$.
Which statement is true for this line?
- (A) Its gradient = $-\frac{1}{2}$ and its y -intercept = 3
(B) Its gradient = 2 and its y -intercept = -3
(C) Its gradient = 2 and its y -intercept = 3
(D) Its gradient = $\frac{1}{2}$ and its y -intercept = 3

4. Tommy is drawing the graph $y = 5 - 2x$. Which sketch could be the correct graph?

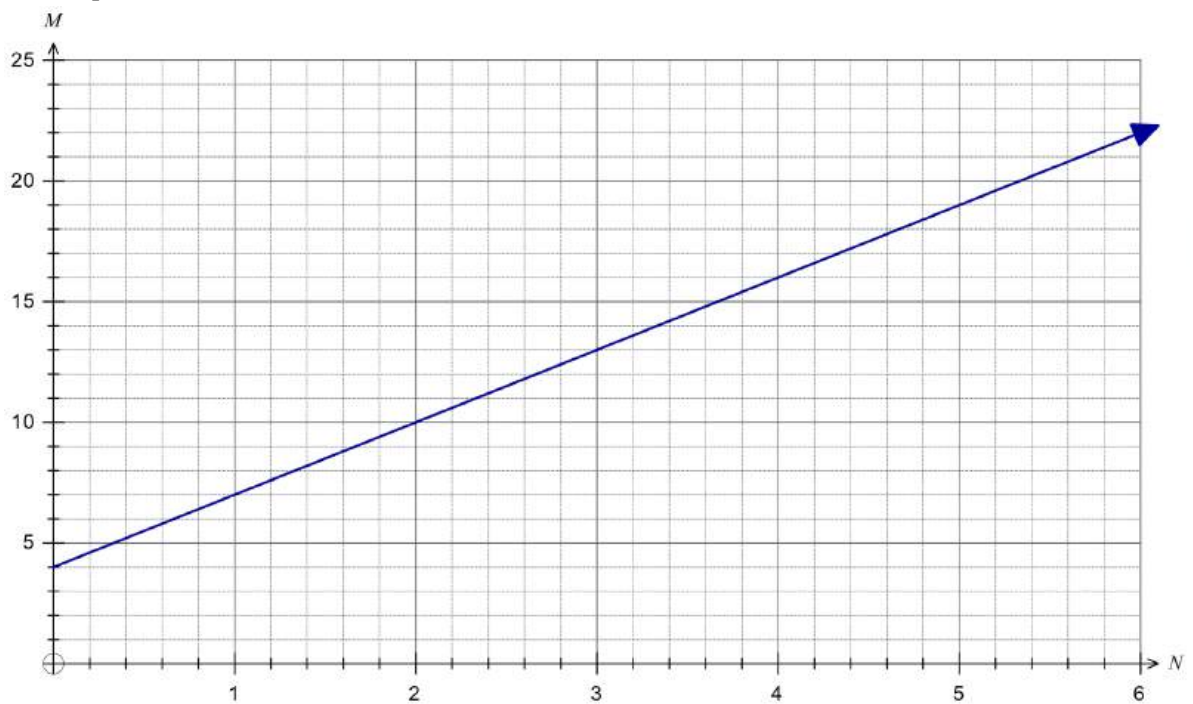


5. Two quantities V and W have a direct variation relationship and $W = 92.0$ when $V = 36.8$. Which equation could describe the relationship?

- (A) $W = 1.5V$
(B) $W = 2.5V$
(C) $W = 5.0V$
(D) $W = 55.2V$

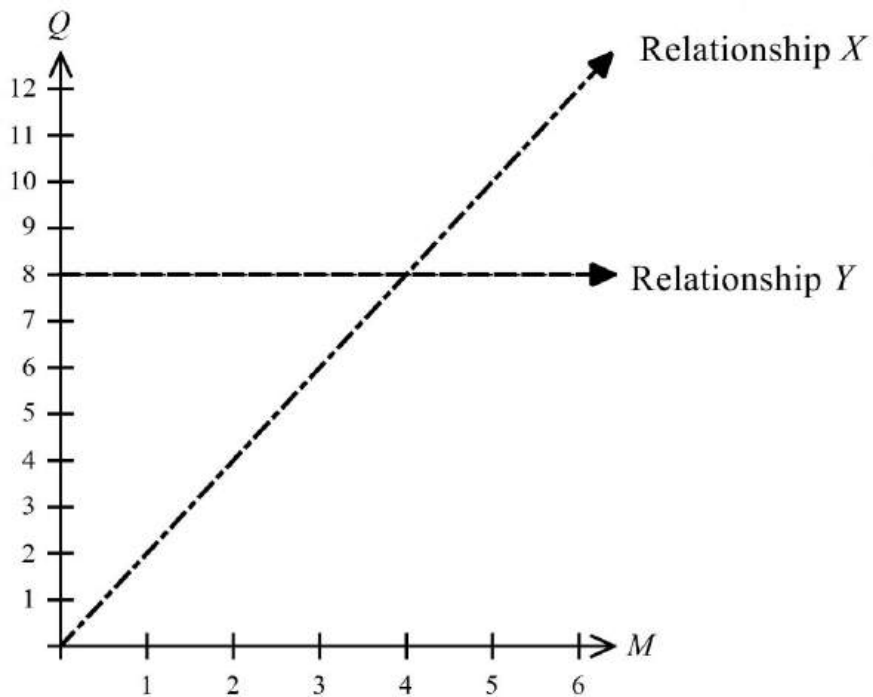
6. The graph below shows the straight-line relationship between two variables M and N .

What is equation of the line?



- (A) $M = 3N + 4$
(B) $M = 0.6N + 4$
(C) $M = 4N + 0.6$
(D) $M = 4N + 3$

7. The graph below shows two lines which represent two different relationships between quantities M and Q .

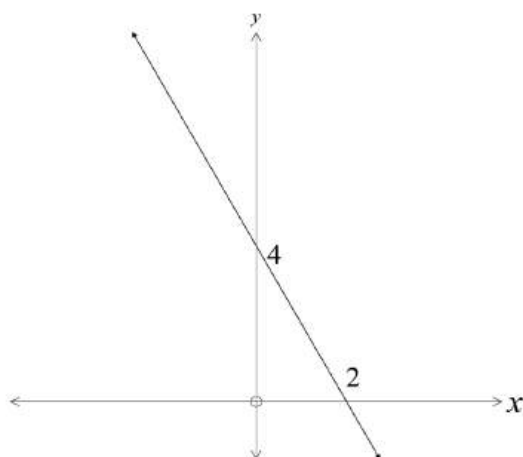


Which relationship is an example of direct variation?

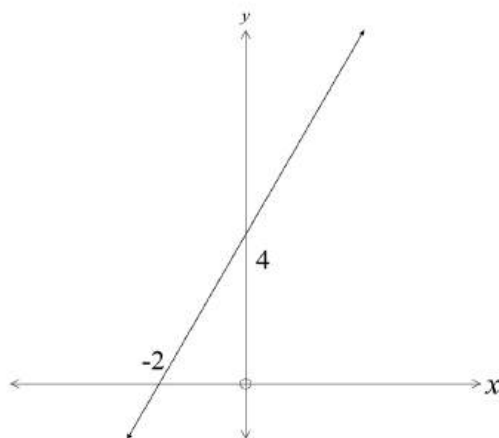
- (A) Both relationships.
- (B) Neither relationship.
- (C) Relationship X only.
- (D) Relationship Y only.

8. Which is the graph of the line $y = 2x - 4$?

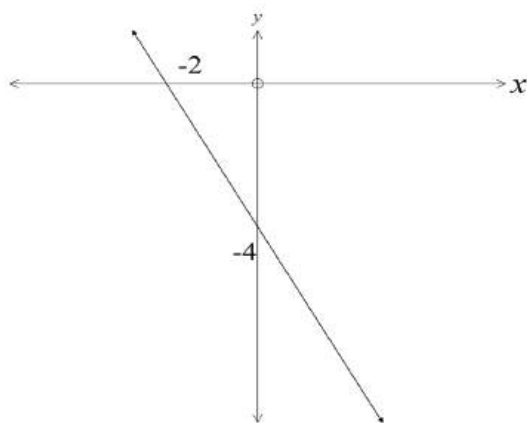
(A)



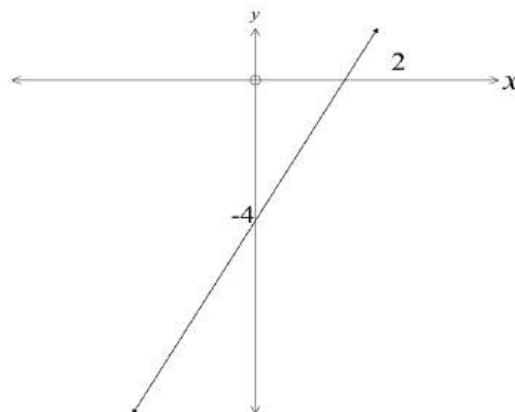
(B)



(C)



(D)



9. A line has a gradient of -3 and a y -intercept of 4 . What is its equation?

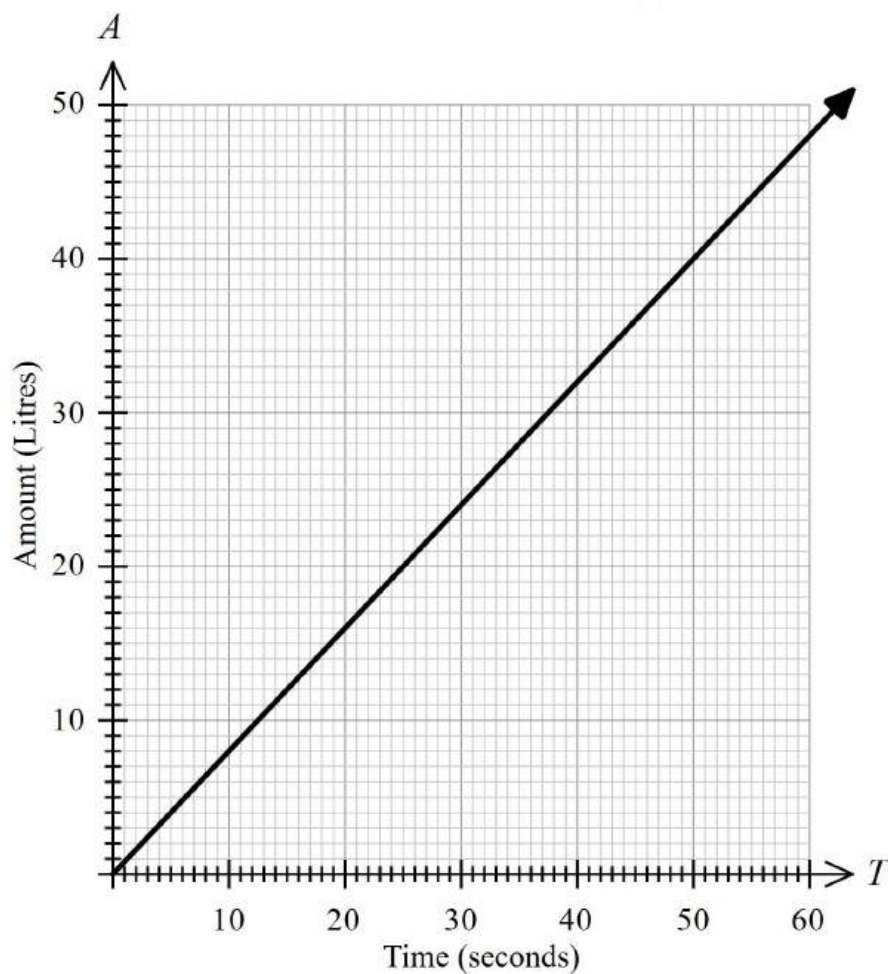
(A) $y = 4 - 3x$

(B) $y = 4x - 3$

(C) $y = 3 - 4x$

(D) $y = 3x - 4$

10. The graph below shows the number of litres of fuel delivered into a car by a pump in a given time up to one minute.



How long would this pump take to fill a tank which has a capacity of 80 litres?

- (A) 1 minute and 20 seconds
- (B) 1 minute and 30 seconds
- (C) 1 minute and 40 seconds
- (D) 2 minutes

Western Mathematics

Mathematics Standard 2

MS-A2 Linear Relationships

REVISION QUESTIONS - Section II

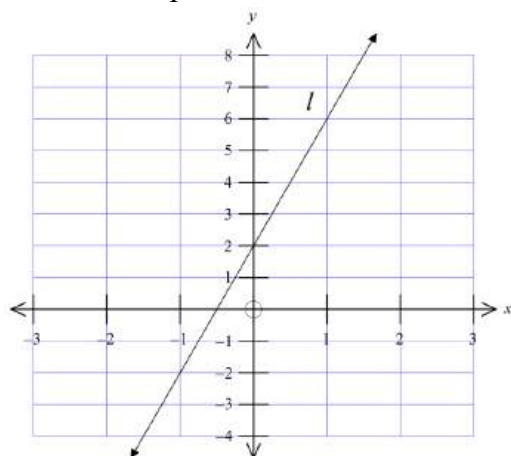
Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (3 marks)

Marks

A line l is shown on the number plane below.



(a) What is the gradient of the line?

1

.....

(b) What is the equation of the line?

1

.....

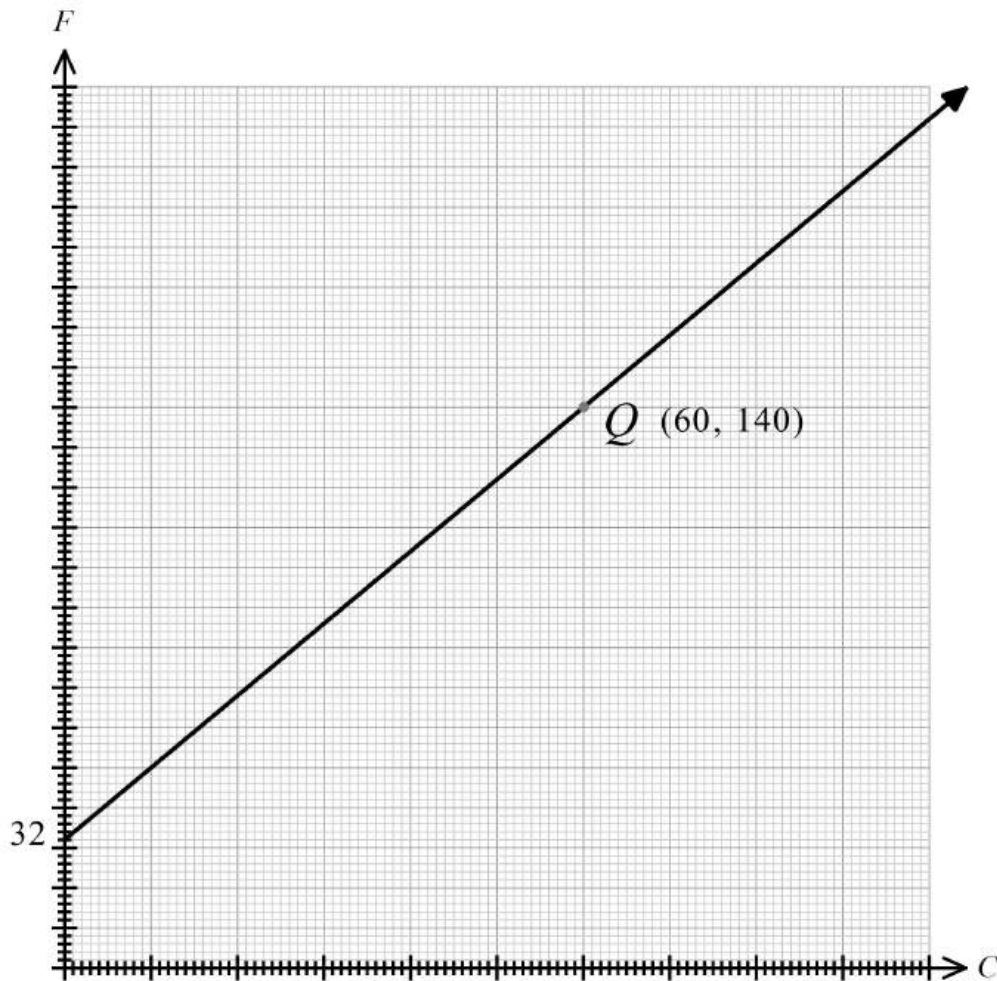
(c) What is the y -value of the point on the line where $x = 15$?

1

.....

Question 12 (3 marks)

The graph below show the line which is used for conversion between temperatures in $^{\circ}\text{C}$ and in $^{\circ}\text{F}$. The point Q (60, 140) lies on the line, and the line crosses the vertical axis at 32.



- (a) What is the gradient of the line?

1

.....
.....

- (b) Give the equation of the line.

1

.....
.....

- (c) Convert a temperature of 150°C to $^{\circ}\text{F}$.

1

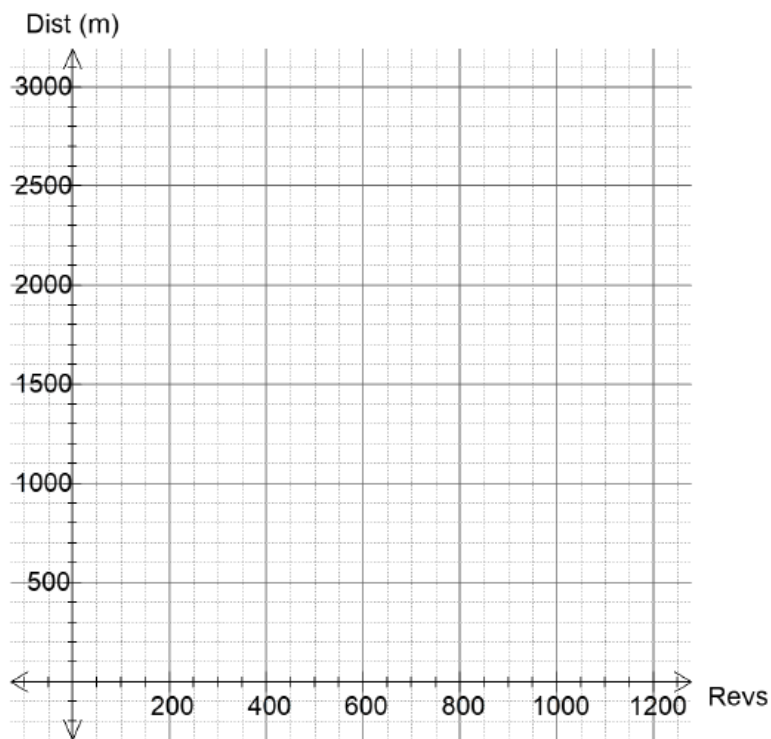
.....
.....

Question 13 (4 marks)

The number of revolutions made by a wheel on a car varies directly with the distance the car travels.

When a wheel on a certain car makes 600 revolutions, the car travels 1 500 m.

- (a) On the grid provided below, draw the line representing this variation.

1

- (b) What is the gradient of this line?

1

.....
.....
.....

- (c) Using D for distance and R for number of revolutions, give the equation of the line.

1

.....
.....

- (d) If the wheel made 2 000 revolutions, what distance would the car travel?

1

.....
.....

Question 14 (3 marks)

The cost of fencing a square field varies directly with its side length.

A square field with a side length of 30 m costs \$1500 to fence.

- (a) Complete the formula $C = k \times \underline{\hspace{2cm}}$ **1**

.....

- (b) Find the value of k . **1**

.....

.....

.....

- (c) Find the cost of fencing a square field with side length = 77 m. **1**

.....

.....

.....

Question 15 (3 marks)

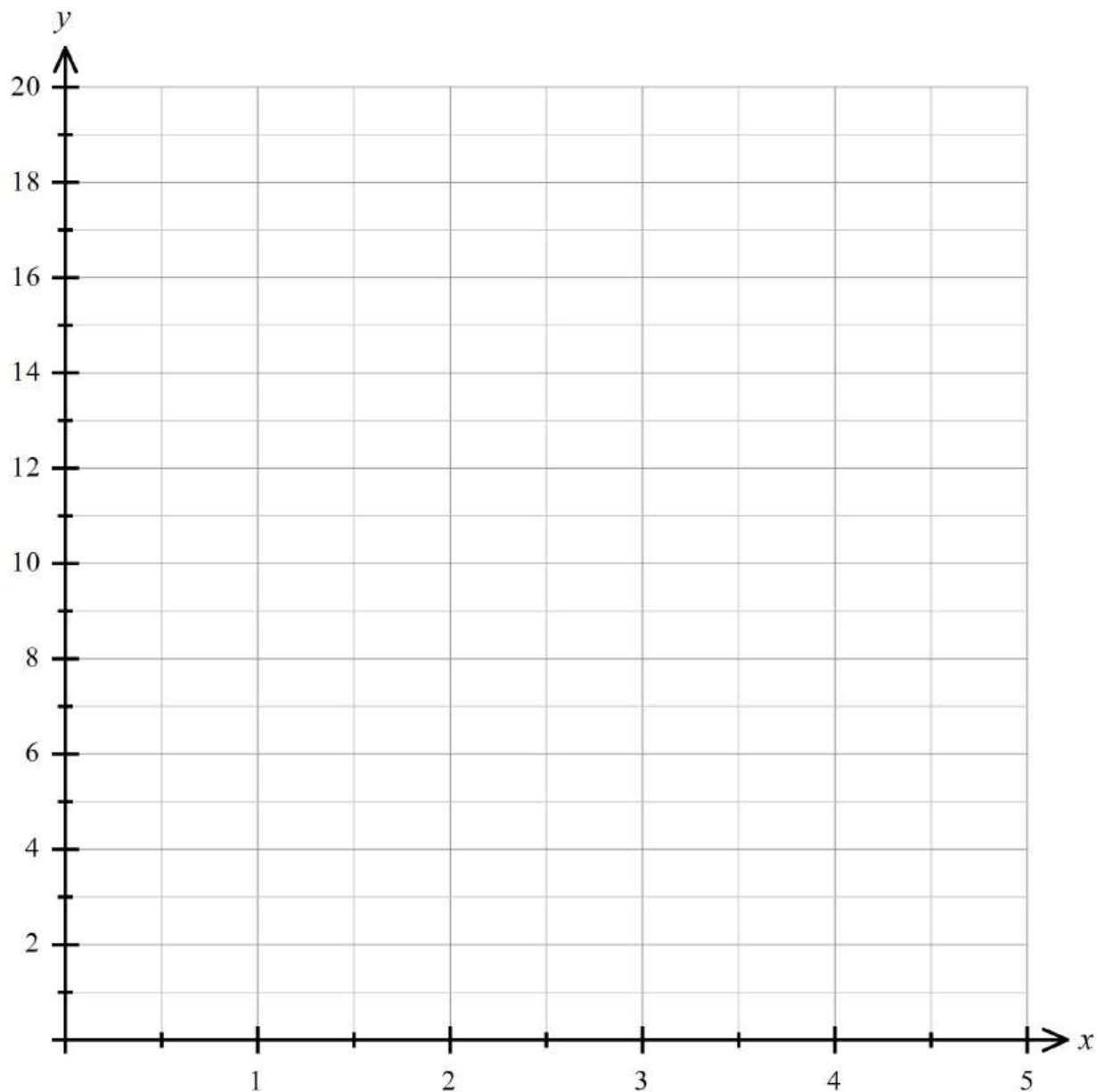
On the number plane below, draw neat sketches of the lines:

3

$$y = 4x + 5 \quad \text{and}$$

$$y = -3x + 12$$

And find the co-ordinates of their point of intersection.



.....

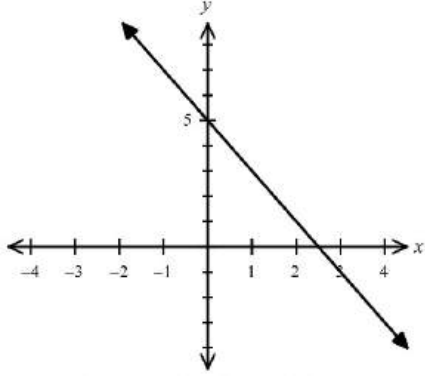
.....

.....

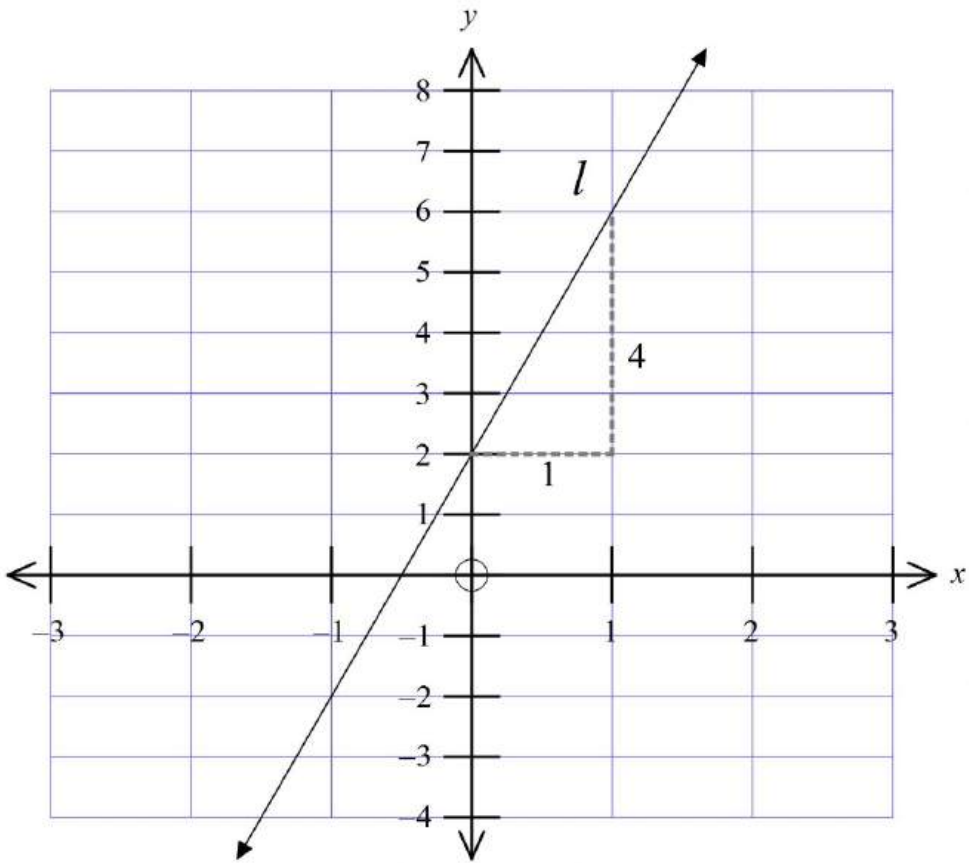
End of Revision

Western Mathematics MS-A2 Linear Relationships
Section I - Worked Solutions

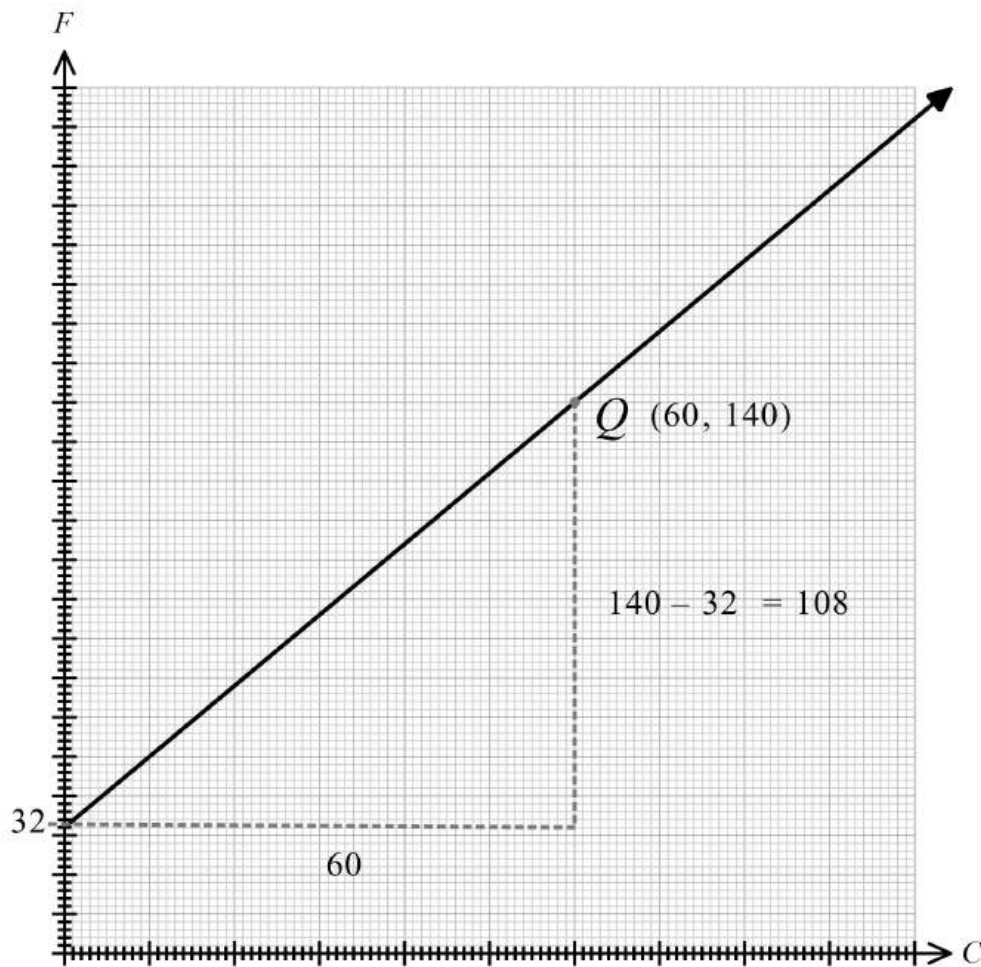
Revision

No	Working	Answer
1	$m = \frac{\text{rise}}{\text{run}}$ $= -\frac{6}{2}$ $= -3 \quad (\text{It is negative as the line is falling from left to right.})$	B
2	$m = \frac{\text{rise}}{\text{run}}$ $= \frac{12}{4}$ $= 3 \quad (\text{It is positive as the line is rising as you go from left to right.})$	C
3	$x - 2y + 6 = 0$ $2y = x + 6$ $y = \frac{1}{2}x + 3$ $\text{Gradient} = \frac{1}{2}$ $y - \text{intercept} = 3$	D
4	$y = 5 - 2x$ $y = -2x + 5$ $m = -2$ $b = 5$ Passes through 5 on y axis gradient = -2, so for each 1 unit across, go down 2 units 	A
5	Because they are in direct variation, we can say that $W = KV \text{ where } K \text{ is the constant of variation}$ So $92.0 = K \times 36.8$ $K = \frac{92.0}{36.8}$ $K = 2.5$ $W = 2.5V$	B

6	Considering points (0, 4) and (2,10) $m = \frac{\text{rise}}{\text{run}}$ $= \frac{6}{2}$ $= 3$ y- intercept = 4 Equation $y = mx + b$ $M = 3N + 4$	A
7	Only relationship X passes through the origin and is a straight line, so it is the only example of direct variation.	C
8	Graph D has a gradient of 2 and y-intercept of -4 so has equation $y = 2x - 4$	D
9	$y = mx + c$ $y = -3x + 4$ $y = 4 - 3x$	A
10	The graph is a direct variation and so $A = kT$ and $A = 40$ when $T = 50$ (or other points on the graph) $40 = k \times 50$ $k = \frac{40}{50} = 0.8 = \frac{4}{5}$ $A = 0.8T$ $80 = 0.8 \times T$ $T = \frac{80}{0.8} = 100 \text{ seconds}$ $= 1 \text{ minute and } 40 \text{ seconds}$	C

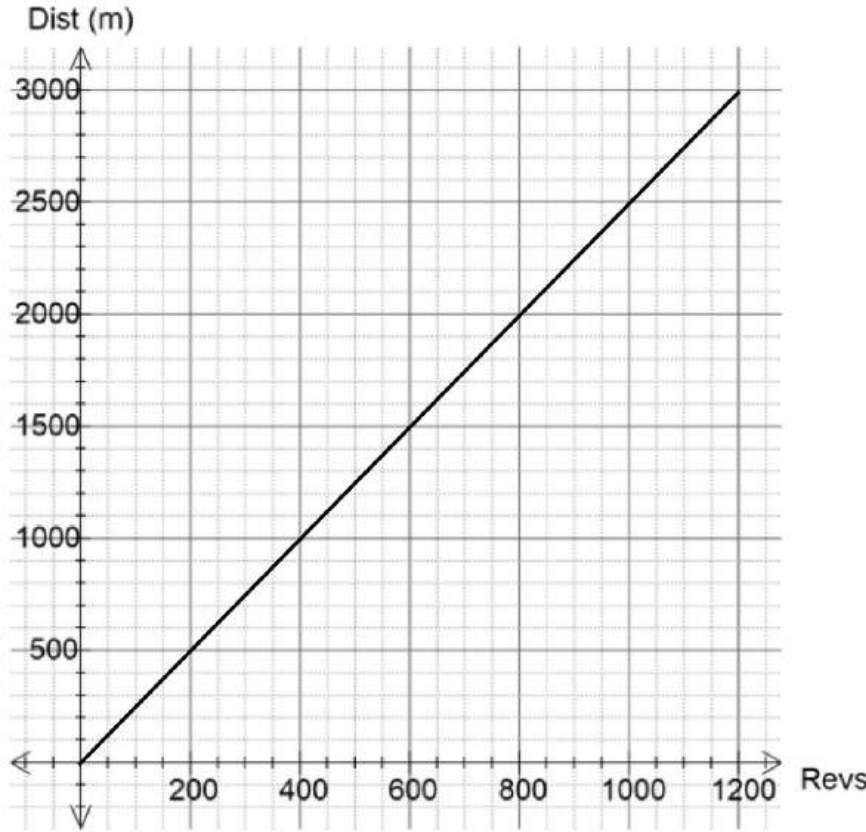
	MS-A2 Linear Relationships Section II Solution	Marks
Question 11		
(a)	 gradient = $m = \frac{4}{1} = 4$	1
(b)	gradient = $m = \frac{4}{1} = 4$ y – intercept = $c = 1$ equation $y = mx + c$ $y = 4x + 1$	1
(c)	When $x = 15$ $y = 4 \times 15 + 1 = 61$	1

Question 12



(a)	Gradient = $m = \frac{108}{60} = 1.8$	1
(b)	y – intercept = $c = 32$ equation $y = mx + c$ $F = 1.8C + 32$	1
(c)	When $C = 150$ $F = 1.8 \times 150 + 32$ $= 302^\circ F$	1

Question 13

(a)		1
(b)	$\text{Gradient} = \frac{\text{rise}}{\text{run}}$ $= \frac{1500}{600}$ $= 2.5$	1
(c)	Variation has form $y = mx$ $D = 2.5R$	1
(d)	$D = 2.5R$ $D = 2.5 \times 2000$ $= 5\,000 \text{ m}$	1

Question 14

(a)	$C = k \times L$ (L = side length)	1
(b)	$C = k \times L$ $1500 = k \times 30$ $k = \frac{1500}{30} = 50$	1
(c)	$C = 50L$ $= 50 \times 77$ $= \$3850$	1

Question 15